ANUP SINGH
230101053

$$I = \begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

$$I_x = \begin{bmatrix} -10 & 0 & 30 \\ -40 & 0 & 80 \\ -30 & 0 & 50 \end{bmatrix} \qquad I_y = \begin{bmatrix} -10 & -40 & -30 \\ 0 & 0 & 0 \\ 30 & 80 & 50 \end{bmatrix}$$

$I_x = 80$ $\qquad$ $I_y = 80$

$$R = \det(M) - k(\text{trace}(M^2))$$

$$\det(M) = \begin{bmatrix} (80)^2 & 6400 \\ 6400 & (80)^2 \end{bmatrix}$$

$= 6400 - 6400$

$\Delta = 0$

$$R = 0 - k(\text{trace}(M)^2)$$

$0 - 0.04 \times (160)^2$

$= -1024$

$R < 0$